

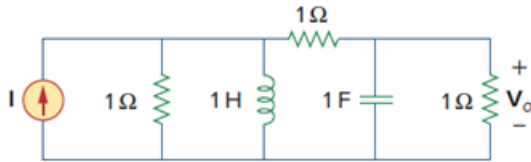
Problem

For the network illustrated in Fig. 14.85, find

(a) the transfer function $\mathbf{H}(\omega) = \mathbf{V}_o(\omega)/\mathbf{I}(\omega)$,

(b) the magnitude of $\mathbf{H}$ at $\omega_0 = 1$ rad/s.

Figure 14.85:



Step-by-step solution

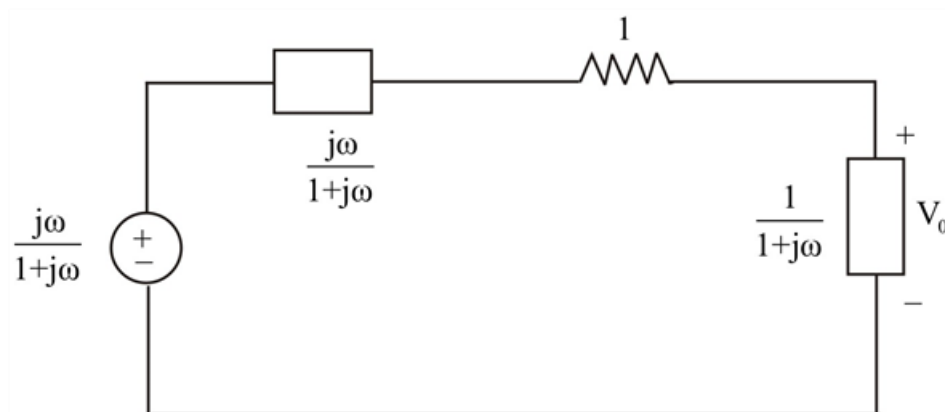
Step 1 of 4

$$(a) \quad 1 \parallel j\omega = \frac{j\omega}{1+j\omega}$$

$$1 \parallel \frac{1}{j\omega} = \frac{\frac{1}{j\omega}}{1 + \frac{1}{j\omega}} = \frac{1}{1+j\omega}$$

Transform the current source gives the circuit below

Step 2 of 4



Step 3 of 4

$$V_o = \frac{\frac{1}{1+j\omega}}{1 + \frac{1}{1+j\omega} + \frac{j\omega}{1+j\omega}} \cdot \frac{j\omega}{1+j\omega} I$$

$$H(\omega) = \frac{V_o}{I} = \frac{j\omega}{2(1+j\omega)^2}$$

Step 4 of 4

$$(b) \quad H(1) = \frac{1}{2(1+j)^2}$$

$$|h(1)| = \frac{1}{2(\sqrt{2})^2} = 0.25$$